

Research Method of Data Science

Week 4: Literature Synthesis and the Research Proposal

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Welcome to Week 4!

Literature Synthesis and the Research Proposal

This week's output: A draft literature review with a supported gap statement and a short self-review.



Agenda

Week 4 · Tuesday meeting

1. Organizing the literature review
 2. Writing synthesis and building an argument
 3. Research gaps and conceptual frameworks
 4. Writing workflow: Word and Zotero
 5. Apply the lesson and prepare for Thursday
- Appendix: writing checklist and software instructions

Learning objectives

By the end of this week, you should be able to:

1. organize literature around meaningful themes or debates;
2. synthesize evidence across sources rather than list summaries;
3. distinguish a defensible research gap from a simple omission;
4. sketch a provisional conceptual framework;
5. connect the literature review to the proposed study;
6. revise writing using structured self and peer feedback.

The literature review in your final project

Your final individual paper includes:

Introduction · Literature review · Data and methods · Results · Discussion · Conclusion · Limitations

Week 3 helped you find and evaluate sources.

This week, you will organize those sources into a **written argument that explains why your study is needed.**

Opening question: What did your reading change about your research question?

The background of the slide features a large, faint, circular watermark of the Seoul National University (SNU) seal. The seal contains the university's name in Korean and English, the year 1960, and the letters IHS.

1. Organizing the literature review

Non-organized literature reviews

A non-organized literature review simply lists sources without connecting them to a central argument or theme.

How doing a literature review feels like



DRFREDPHD

Me in the middle of my literature review



An evidence matrix and a literature map

An **evidence matrix** records what each source contributes, for example:

Source	Contribution
A	Shows the frequency of AI use
B	Reports on the impact of AI on learning outcomes
C	Provides insights into student attitudes towards AI

A **literature map** shows how sources connect through themes, debates, or explanations.

Theme	Sources	Connection to the research question
What counts as learning?	A and B	Assisted scores and independent ability are different outcomes
How do students use AI?	A and C	Frequency of use may conceal differences in purpose

Fictional teaching example. A source may appear in more than one theme.

Before the challenge: what are we organizing?

Guiding research question:

How might different ways of using generative AI relate to university students' independent learning?

A **theme** is a shared issue that several sources help us understand.

For example, a review of online education might group sources under:

- **access to learning;**
- **student engagement.**

A theme should help answer the research question. It can bring together studies using different methods.

Your task: organize sources so the reader can follow an argument.

Pair challenge: organize six source notes – 5 minutes

Question: How might different ways of using AI relate to independent learning?

Fictional source notes describe each study's focus, not its findings.

Source	What the study examines
A	How often students use AI and how they rate their own learning
B	Students' task scores while AI assistance is available
C	Interview accounts of how students use AI when they encounter difficulty
D	Students' performance on a later task completed without AI
E	Whether the relationship between AI use and learning differs by students' prior preparation
F	Differences between using AI to check ideas and using it to replace one's own effort

Your task: build a two-theme outline

In pairs, work for three minutes:

1. Name **two themes** that help answer the guiding question.
2. Place sources **A–F** under them. A source may support both.
3. Choose which theme the reader should encounter first.
4. Complete this sentence:

“We begin with ___ because the reader needs to understand ___ before considering ___.”

Then: two pairs present for one minute each.

Different maps are acceptable when you explain their logic.

Themes and debates give sections a purpose

A theme groups studies around a shared issue.

Example: how studies define and measure learning.

A debate concerns competing interpretations or explanations.

Example: whether assistance supports practice or replaces it.

A useful heading tells the reader what the section examines:

| “Assisted performance and independent learning”

Each section should help answer a question relevant to your study.

The review needs an overall argument

A possible outline for the teaching example:

1. **Opening:** define the question and scope of the review.
2. **Theme 1:** compare what studies count as learning.
3. **Theme 2:** examine different purposes of AI use.
4. **Closing:** identify what remains unresolved and explain the study's focus.

Why this order? The reader first needs to understand the outcome before interpreting evidence about AI use.

2. Writing synthesis and building an argument



Three sources, three kinds of evidence

Invented examples for practice, not published findings

Source	Evidence	Finding in the example
A	Survey of use frequency and self-rated thinking	More use accompanies lower self-ratings
B	Experiment measuring an assisted task	AI access improves immediate scores
C	Interviews about learning practices	Students describe both checking ideas and replacing effort

Can these sources establish whether AI improves later independent learning?

The job of a synthesis paragraph

A paragraph can make four moves:

1. **Claim:** state the idea the paragraph develops.
2. **Evidence:** bring relevant sources into the discussion.
3. **Interpretation:** explain agreement, differences, or limits.
4. **Link:** show what follows for the next paragraph or your study.

These are writing moves, not a compulsory four-sentence formula.

A worked synthesis paragraph

Evidence about AI and learning depends on the outcome measured. In these fictional examples, A associates frequent use with lower self-rated thinking, while B reports higher scores during an assisted task. The findings need not conflict: self-perceived ability and assisted performance capture different outcomes. Neither establishes how students perform later without assistance. This distinction motivates closer attention to independent learning and to the ways students use AI.

Pair annotation — 4 minutes

Label each sentence: **C**laim, **E**vidence, **I**nterpretation, or **L**ink.

Then decide: **Which sentence would be most damaging to delete, and why?**

Pair exercise: a second connected paragraph

Use fictional source C alongside A or B.

In **three minutes**, draft two or three sentences that:

- begin with differences in the **purpose of AI use**;
- compare evidence from at least two sources;
- connect to the preceding paragraph about outcomes.

Two pairs will read their versions. Explain the connection you made.

Transitions explain a relationship

Weak connection

“Another study examined students' AI use.”

More informative connection

“While the preceding studies distinguish learning outcomes, the next group examines how students use AI during a task.”

Useful relationships include **contrast, extension, explanation, and qualification**.

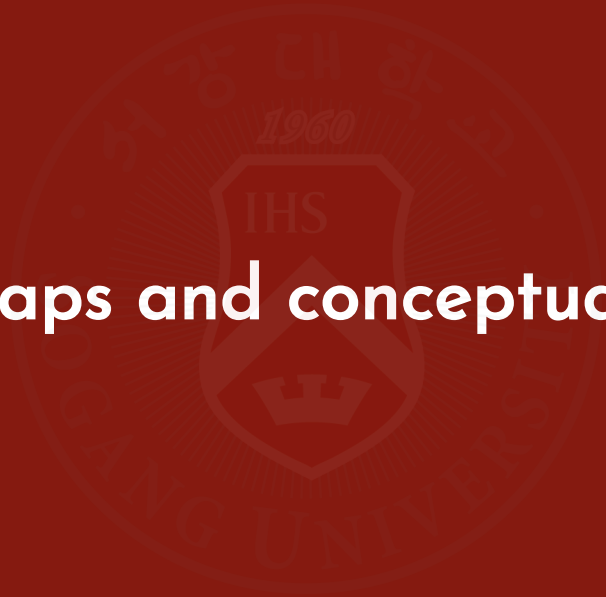
“Furthermore” alone does not explain why the next study belongs.

Critical writing matches claims to evidence

Overstatement	More defensible wording
"A proves that AI reduces thinking."	"A reports an association with self-rated thinking."
"B contradicts A."	"B examines a different outcome under a different design."
"These studies are weak."	"These measures cannot directly answer the question about later unaided performance."

Cite the sentence's actual support. Use quotation marks and a locator for direct quotations.

3. Research gaps and conceptual frameworks



It's not a trivial task to identify a gap

**ME TRYING TO FIND GAP
FROM LITERATURE REVIEW**



A gap grows out of the synthesis

A gap statement explains:

- **What the reviewed evidence establishes**
- **What remains unresolved**
- **Why resolving it matters**

Example: “The selected studies distinguish outcomes and uses of AI but leave their connection to independent performance unresolved. Clarifying that connection would help distinguish immediate assistance from learning.”

This is a claim about **these selected sources**, pending a broader check.

Gap triage – 6 minutes

Classify each claim as **simple omission**, **promising but unverified**, or **provisionally defensible**.

A. “None of my five papers studies Korea.”

B. “The selected papers measure frequency but do not distinguish purposes of use, which could change how their results are interpreted.”

C. “Across the databases and terms documented in our search, studies distinguish assisted scores from self-reports, but none directly evaluates later unaided performance. This leaves retention unresolved.”

For each: What further evidence could strengthen—or overturn—the claim?

The framework expresses what the review suggests

A **conceptual framework** identifies the concepts and connections guiding your study.

Concept	Meaning in this example	Possible connection
Purpose of AI use	Checking ideas or replacing effort	Different practices may relate differently to learning
Independent learning	Performance without AI assistance	Distinct from success while help is available
Prior preparation	Knowledge before the task	May relate to both use and performance

These are **proposed connections**, not established causal effects.

Framework sketch – 5 minutes

On paper, draw three boxes:

1. your focal concept or outcome;
2. one possible explanatory concept;
3. one contextual, moderating, or confounding concept.

Add arrows. Under each arrow, write:

| “I think these may be connected because ...”

Mark every connection **provisional** and identify which source or additional evidence could support it.

The closing paragraph positions your study

A review's ending should connect:

The evidence reviewed, the remaining question, and your study's focus.

“Taken together, the reviewed evidence suggests [supported conclusion]. However, [specific uncertainty] remains. The present study therefore examines [focused question] in [defined context].”

Your final project will answer the question through evidence and analysis. The literature review explains why the question deserves attention.

4. Writing workflow: MS Word and Zotero

The writing argument stays the same

Tool	Role
Zotero	Organize sources and maintain citation metadata
MS Word	Write and format your document directly

Use **Word + Zotero** in the live demonstration. Students who prefer LaTeX may use the documented Overleaf route in the Appendix.

The software manages formatting—not the accuracy of your interpretation.

MS Word and Zotero demonstration

1. Write a paragraph under a meaningful heading.
2. Use **Add/Edit Citation** in Word's Zotero tab.
3. Insert the bibliography and check the details.
4. Correct metadata in Zotero and **Refresh** in Word.

Save the editable **.docx** and export a PDF.

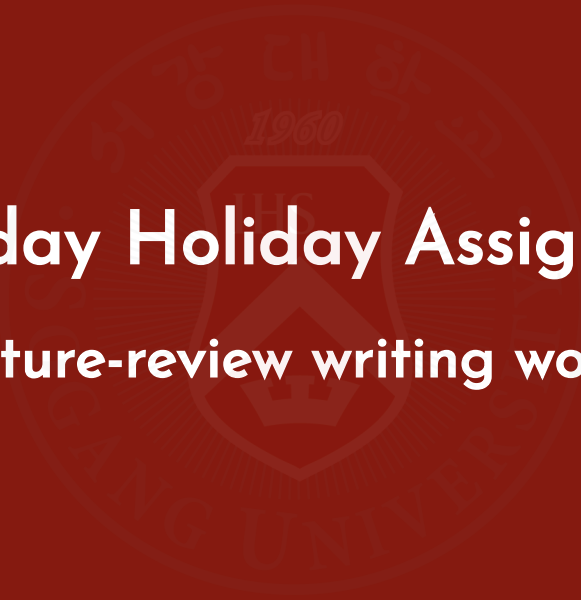
The bibliography helps readers locate the evidence. It cannot check whether your interpretation is accurate.

5. Apply the lesson and prepare for Thursday

For your own project, write:

1. one meaningful theme heading;
2. one relationship you need to explain across sources;
3. one possible gap, labelled **omission**, **unverified**, or **provisionally defensible**;
4. one concept that belongs in your framework.

Have your Week 3 matrix, verified sources, editable draft, and Zotero library available for the Thursday assignment.

The seal of Chung-Ang University is a circular emblem. It features a central shield with a crown on top. The shield is divided into four quadrants, each containing a different symbol. The year '1960' is inscribed above the shield. The Korean text '중앙대학교' (Chung-Ang University) is written in a circular path around the shield. The English text 'CHUNG-ANG UNIVERSITY' is written in a circular path around the bottom half of the shield.

Thursday Holiday Assignment

Individual literature-review writing workshop at home

Assignment output

Complete independently during the Thursday holiday:

1. a literature map with two themes and at least three verified sources;
2. two connected synthesis paragraphs—about 200–300 words;
3. a supported provisional gap statement—about 60–100 words;
4. one substantive revision and a short self-review;
5. checked citations, references, and an AI-use statement.

This remains an individual component of your final research project.

A manageable 75-minute sequence

Minutes	Individual task
10	Map at least three verified sources into two themes
30	Write or revise two connected paragraphs—200–300 words total
10	Write a 60–100-word provisional gap statement
10	Use the self-review checklist and revise one weak passage
10	Check citations and references; format the document
5	Export the PDF, add the AI-use statement, and submit

Reuse your Week 3 paragraph where useful. This is structured individual work, not an additional 75 minutes beyond the assignment.

Individual self-review

Label the sentences in your paragraphs:

- **C** — claim;
- **E** — evidence;
- **I** — interpretation;
- **L** — link.

Then ask:

1. Does each theme have a purpose in the argument?
2. Does at least one paragraph directly compare two sources?
3. Does the transition explain a real relationship?
4. Does the gap follow from the evidence, and why does it matter?
5. What evidence could weaken or overturn the gap claim?

Identify one weakness and make one substantive revision.

Final verification

- Every citation supports the claim attached to it.
- Authors' findings are distinguishable from your interpretation.
- Every in-text citation appears in the references.
- The transition explains a real relationship.
- The gap is labelled appropriately: **omission**, **unverified**, or **provisionally defensible**.
- The document contains no invented findings or citations.
- The document includes an AI-use statement, even if no AI was used.

Use real citations for your selected sources. Do not cite the fictional classroom examples.

Submission and next class

Upload to **Cyber Campus**, by the announced deadline:

- one PDF containing your question, literature map, synthesis paragraphs, gap, references, and short self-review;
- the editable **.docx** or Overleaf source ZIP.

At the next live class, bring the draft and current question. We will begin with a brief peer exchange before qualitative research.

Prepare to ask a peer for one strength in the synthesis and one specific improvement to the reasoning or gap.

Also consider:

| What could interviews, observations, or documents help you understand about your question?

Next topic: **qualitative research**.

The background of the slide features a large, faint, circular seal of Yonsei University. The seal contains the university's name in Korean and English, the year 1960, and a central shield with a crown and the letters 'IHS'.

Appendix

Writing checks and software reference

A literature map you can use

Theme or debate	Supporting sources	Connection	Role in your argument
Theme 1	Actual author/date citations	Agreement, contrast, or complementary evidence	What does the reader learn?
Theme 2	Actual author/date citations	How do methods or contexts differ?	How does this prepare the gap?

A source can support both themes.

The table records the relationships you will explain in prose.

Self-review and later peer feedback

1. What is the main claim of each paragraph?
2. Where does the writer compare sources?
3. Does each citation support the claim attached to it?
4. Does the paragraph order make sense?
5. Does the gap follow from the evidence, and why does it matter?

At home: identify one weakness and revise it.

Next live class: ask a peer to identify one strength and one specific improvement.

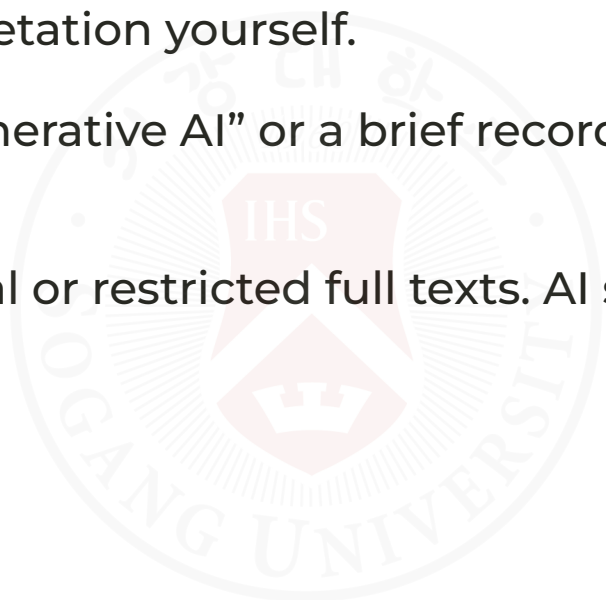
Responsible AI assistance

You may use AI to suggest organization or critique clarity from **your verified notes**, subject to course rules.

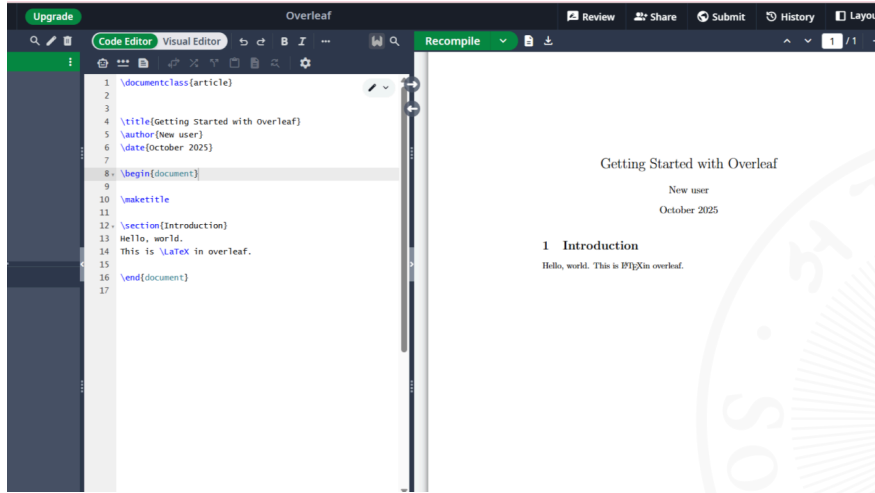
Check every source and interpretation yourself.

Include either “I did not use generative AI” or a brief record of the tool, task, output used, and verification.

Do not upload sensitive material or restricted full texts. AI suggestions are not evidence of a research gap.



Optional route: Overleaf



Overleaf is a browser-based LaTeX editor that generates a formatted PDF from source files.

Use it only if you prefer this workflow. Word + Zotero is fully acceptable for the course.

LaTeX uses commands to describe document structure. The manual workflow below does not require paid Zotero integration.

Overleaf workflow

1. Open a blank project and locate `main.tex`.
2. Add a title, heading, and paragraph using the following appendix example.
3. Create or upload `references.bib`.
4. Use the exact BibTeX citation key in `\citep{key}`.
5. Select **Recompile** and inspect the citation and bibliography.
6. Download the PDF and source ZIP for submission.

`main.tex` contains your writing.

`references.bib` contains bibliographic records.

Zotero to Overleaf without paid integration

1. Correct and verify the source metadata in Zotero.
2. Export selected references as **BibTeX**.
3. Save the export as `references.bib`, without attached PDFs.
4. Upload the file to Overleaf.
5. Use each entry's exact citation key in the text.

This route does **not** update automatically. After correcting Zotero metadata, export and upload the file again.

See [Zotero exporting](#) and [Overleaf reference-manager integration](#).

Overleaf: a minimal working document

Create a blank project. Replace `main.tex` with:

```
\documentclass{article}
\usepackage[authoryear,round]{natbib}
\title{My Literature Review}
\author{Your Name}
\date{}
\begin{document}
\maketitle
\section{Literature review}
A guide to literature reviews is available
\citep{randolph2009}.
\bibliographystyle{plainnat}
\bibliography{references}
\end{document}
```



Overleaf: the bibliography file

Create a new file named `references.bib`. Paste:

```
@article{randolph2009,  
  author = {Randolph, Justus J.},  
  title = {A Guide to Writing the Dissertation  
          Literature Review},  
  journal = {Practical Assessment, Research,  
            and Evaluation},  
  year = {2009},  
  volume = {14},  
  note = {Article 13},  
  doi = {10.7275/b0az-8t74}  
}
```

Recompile. The PDF should show an author-date citation and reference.

Your own references in Overleaf

1. Select your sources in Zotero and export as **BibTeX**.
 2. Save the export as `references.bib`, without attached PDFs.
 3. Upload it, replacing the demonstration bibliography.
 4. Replace the sample citation with your actual citation keys.
 5. Recompile, inspect citations, and download the PDF and source ZIP.
- A key is the label immediately after `{` in a bibliography entry. Keys must match exactly.

Quick troubleshooting

Problem	First check
Zotero tab missing	Use desktop Word and check plugin installation
Citation appears as ?	Match the key to an entry in <code>references.bib</code>
Overleaf compilation fails	Read the first error and check paired braces
Bibliographic detail is wrong	Correct Zotero metadata, then refresh or re-export

Save your work. After five minutes of difficulty, continue in an available editor and record the issue.

References and software help

- Randolph (2009), A Guide to Writing the Dissertation Literature Review
- Zotero: Word plugin
- Zotero: exporting references
- Overleaf: Learn LaTeX in 30 minutes
- Overleaf: citations with natbib
- Overleaf: Zotero integration and manual export

Sources A, B, and C are fictional teaching examples. The writing exercises make no factual claims about AI's effects on learning.